



PROFILE

<https://stefannug.github.io/>

Researcher in formal reasoning with a background in mathematics and computer science. My current research focuses on automated reasoning, SMT solving, and formal verification, particularly satisfiability certification for quantified formulas. More broadly, I am interested in the connections between mathematical logic, theorem proving, and automated methods for solving computational problems. I am pursuing a PhD at ČVUT while working as a researcher at the Institute of Computer Science, Czech Academy of Sciences.

CONTACT

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RESEARCH INTEREST

- Formal Verification
- Automated Reasoning
- Theorem Proving
- Logical Foundations of Computation
- Formalised Mathematics
- Foundations of Mathematics
- Philosophical logic

TECHNICAL SKILLS

- Language: Bahasa Indonesia (mother tongue), English
- Programming: Python (developing), Java, C
- Scientific Computing: Maple, Matlab
- Typesetting: LaTeX

ANGGHA 'STEFAN' NUGRAHA

EDUCATIONS

PhD in Computer Science – Czech Technical University (ČVUT)

2025 - present

- Supervisor: doc. Dipl.Ing. Stefan Ratschan, Dr.-tech
- Research focus: Automated reasoning, SMT solvers, verification

PhD in Mathematics – University of Canterbury (NZ)

2014 - 2019

- Thesis title: Naïve Infinitesimal Analysis.
- Supervisor: Dr. Maarten McKubre-Jordens and Dr. Hannes Diener.

Master of Computer Science – University of Indonesia (IDN)

2011 – 2012

- Thesis title: Verification of Square Matrix Multiplication Algorithm using Hoare Logic.
- Supervisor: Dr. Ade Azurat

Bachelor of Science in Mathematics – University of Indonesia (IDN)

2006 – 2009

- Thesis title: Generating Function Application to DNA Sequencing.
- Supervisor: Dr. Kiki A. Sugeng and Dr. Denny R. Silaban

PUBLICATIONS

Anggaha Nugraha. [*Paraconsistent Dominated Convergence*](#). arXiv:2607.15723 [math.LO], **2026**.

Anggaha Nugraha. [*A Constructive Field of Infinitesimals: Chunk and Permeate Approach*](#). arXiv:2607.08206 [math.LO], **2026**.

*This is an updated, comprehensive version that expands upon and supersedes the 2020 preprint (arXiv:2009.11424) and PhD thesis.**

Stefan Ratschan, Anggaha Nugraha, Mikoláš Janota, Marek Dančo. [*Inductive Satisfiability Certification for Universal Quantifiers and Uninterpreted Function Symbols*](#). arXiv:2602.16335 [cs.LO], **2026**.

Anggaha Nugraha, M. McKubre-Jordens, and H. Diener. [*Naïve Infinitesimal Analysis: Its Construction and Its Properties*](#). In arXiv.2009.11424 [math.LO], **2020**.

T. Antaresti, Anggaha Nugraha, et al. [*Wavelet denoising and fractal feature selection for classifying simulated earthquake signals from*](#)

*last updated: July 2026

mobile phone accelerometers. In IEEE 11th International Multi-Conference on Systems, Signals & Devices (SSD 14), **2014**.

M. Arzaki and Anggha S. Nugraha. *Translation of an Allen's Temporal Logic Variant to RPNL*. In Journal of Mathematical and Computational Science (3), **2013**, pages 1405-1429.

A. S. Nugraha and T. Basaruddin. *Analysis and Comparison of QR Decomposition Algorithms for Some Types of Matrix*. In Proceedings of Federated Conference on Computer Science and Information System (FedCSIS), **2012**, pages 561-565.

WORKS

Vědecký asistent (Senior Research Assistant)

Institute of Computer Science, Czech Academy of Sciences, Prague

September 8, 2025 - present

Member of the Department of Artificial Intelligence. Research involving automated reasoning, SMT solvers, and AI methods. Currently pursuing a second PhD in related areas.

Lecturer (Assistant Professorial Fellow 5)

Institute of Mathematics, The University of Philippines - Diliman

August 01, 2024 - July 31, 2025

Deliver undergraduate lectures and exams; prepare teaching materials.

Career Interruption: Religious Vocation and Monastic Formation

2019–2024

Academic career break devoted to religious vocation and monastic formation.

Tutor

School of Mathematics and Statistics University of Canterbury, NZ

2015–2018

Leading tutorial classes (about 20 students in one class); marking exams; proctor for quizzes and exams.

Lecturer and Researcher

Faculty of Computer Science University of Indonesia

2012–2014

Preparing and giving classes on the subjects assigned by the Dean; making, giving, and marking exams; managing some teaching assistants; member of Formal Methods in Software Engineering lab.

Teaching Assistant

Faculty of Computer Science University of Indonesia

2011–2012

Helping students to understand some courses, e.g. Discrete Mathematics, Calculus, Linear Algebra.

Lab Assistant**Department of Mathematics University of Indonesia**

2007–2009

Helping students to understand some computer languages (C, C++, Java), Matlab, Maple.

Teaching Assistant**Faculty of Economics University of Indonesia &****Department of Mathematics University of Indonesia**

2008

Giving tutorial classes on Discrete Mathematics and Calculus.

CONFERENCES AND TALKS

New Zealand Mathematics and Statistics Postgraduate Conference

Auckland, New Zealand

2014, 2017, and 2018

Various informal talks in The Netherlands (especially at the University of Amsterdam) when doing research collaboration with Prof. Jan Bergstra

The Netherlands

2016

7th Non-Classical Logic: Theory and Applications

Toruń, Poland

2015

1st National Joint Seminar in Mathematics “Mengungkap Keindahan Matematika: National Seminar in Mathematics”

University of Indonesia, Indonesia

The Generating Function Applications for Counting The Number of Pattern Graph from Main Pattern Graph

2010

GRANTS AND SCHOLARSHIPS

LPDP Scholarship**Ministry of Finance of The Republic of Indonesia**

2014-2019

Scholarship to do my PhD study at the University of Canterbury, NZ.

The New Zealand Mathematical Society (NZMS) Travel Grant

2016

Travel grant to do research collaboration with Prof. Jan Bergstra of the University of Amsterdam, The Netherlands.

The New Zealand Mathematical Society (NZMS) Travel Grant

2015

Travel grant to present my paper at “The 7th Conference on Non-Classical Logic: Theory and Application” in Poland.

IMHERE (Indonesia Managing Higher Education for Relevance and Efficiency) Scholarship for Master Study
2011-2012

ORGANISATIONS

Management and Leadership Skills

2011-2014

Member of Formal Method in Software Engineering Lab in the Faculty of Computer Science University of Indonesia

2011-2012

Member of Lembaga Asisten (association of teaching assistants) in the Faculty of Computer Science University of Indonesia.

2009

- Person in charge of Free Medical Service activity held by Catholic Student Association university of Indonesia.
- Person in Charge of Social Visit to Desa Putera Orphanage held by Catholic Student Association Faculty of Mathematics and Natural Sciences University of Indonesia.
- Committee in Indonesia Combinatory Community Gathering.
- Person in charge of Mathematics Team Competition LOGIKA Java-Bali held by the Department of Mathematics University of Indonesia.

REFERENCES

1. doc. Dipl.Ing. Stefan Ratschan, Dr.-tech.
Institute of Computer Science, Czech Academy of Sciences
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2. Dr. Maarten McKubre-Jordens
School of Mathematics and Statistics University of Canterbury
mkm@papanui.school.nz
3. Prof. Dr. Kiki Ariyanti Sugeng
Department of Mathematics University of Indonesia
kiki@sci.ui.ac.id
4. Phillipa Gourdie (Senior Tutor)
School of Mathematics and Statistics University of Canterbury
phillipa.gourdie@canterbury.ac.nz